Snake — Files ki Poori Documentary (Hindi)

Project: Snake (Android + HTML5 game)

Version: 0.1 (skeleton — sirf dhancha, game ka code baad mein)

Yeh document kya hai: is project ki **har file aur har folder** ka kaam, ek jagah, Hindi mein. Naya developer ise padh kar poora project samajh sakta hai.

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0. Sabse pehle: Project aur APK alag cheezein hain

	Project (source)	**APK (banaya hua package)**
Kya hai	Aapka code, files, folders	Install hone wali file
Kahan	Computer / GitHub	Phone
Andar	`.java`, `.kts`, `.xml`, `.html`	`classes.dex`, `resources.arsc`, `lib/`, `META-INF/`
Badal sakte hain?	Haan, jab chahein	Nahi — dobara build karna padta hai

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1. Poora folder structure

```
Snake/
├── .github/
│   └── workflows/
│       └── build-apk.yml          ← GitHub Actions: har push par APK banata hai
├── app/                           ← App module (asli app yahan hai)
│   ├── build.gradle.kts           ← App ki build settings
│   ├── proguard-rules.pro         ← Release build ke rules
│   └── src/
│       └── main/
│           ├── AndroidManifest.xml    ← App ki pehchaan, permissions, screens
│           ├── java/com/snakegame/app/
│           │   └── MainActivity.java  ← App ka pehla (aur abhi akela) screen
│           ├── res/
│           │   └── values/
│           │       ├── strings.xml    ← App ka naam / text
│           │       └── colors.xml     ← Rang
│           └── assets/                ← Raw files (jo code se padhi jaati hain)
│               ├── index.html         ← Abhi ka placeholder page
│               ├── config/
│               │   └── app_config.json ← Game ki settings (board size, features)
│               └── game/              ← Game ka asli code yahan aayega
├── docs/
│   ├── FILES-HINDI.md             ← Yahi document
│   ├── FILE-TREE.txt              ← Sirf folder tree
│   └── ROADMAP.md                 ← Aage kya-kya banega
├── gradle/
│   └── wrapper/
│       ├── gradle-wrapper.jar
```

```
│       └── gradle-wrapper.properties
├── build.gradle.kts                ← Project-level build file
├── settings.gradle.kts             ← Kaunse module hain + repositories
├── gradle.properties               ← Gradle ki settings
├── gradlew / gradlew.bat           ← Gradle chalane wali script (Linux/Mac / Windows)
├── .gitignore                      ← Git ko kaunsi files nahi bhejni
├── LICENSE                         ← MIT license
└── README.md                       ← Project ka parichay
```

2. Root (sabse bahar wali) files

File	Kaam
`settings.gradle.kts`	Batata hai ki project mein kaunse module hain (`:app`) aur libraries kahan se aayengi (Google, Maven Central).
`build.gradle.kts` (root)	Project-level plugins — jaise Android plugin ka version (8.9.3).
`gradle.properties`	Gradle ki settings: memory, caching, AndroidX on/off.
`gradlew` , `gradlew.bat`	Gradle ko chalane wali script. `./gradlew assembleDebug` isi se chalta hai. Ye Gradle ko khud download karti hai.
`gradle/wrapper/*`	Wrapper ki files — batati hain kaunsa Gradle version use hoga (8.11.1).
`.gitignore`	Git in files ko ignore karta hai: `build/` , `.gradle/` , `local.properties` , `*.apk` wagairah.
`LICENSE`	MIT license — code ka istemal karne ki shartein.
`README.md`	Project ka parichay, install steps, links.
`local.properties`	**Is machine par** Android SDK ka path. Ye GitHub par nahi jaati (gitignored), har computer par alag hoti hai.

3. `app/` — app module

File	Kaam
`app/build.gradle.kts`	App ka `namespace` , `applicationId` (play store par permanent ID), `minSdk` 24, `targetSdk` 35, `compileSdk` 36, version, build types.
`app/proguard-rules.pro`	Release build mein code chhota/obfuscate karne ke rules.
`app/src/main/AndroidManifest.xml`	App ki pehchaan: naam, icon, theme, permissions, kaunsi activity pehle khulegi.
`app/src/main/java/…`	Java/Kotlin code — yahan `MainActivity` hai.
`app/src/main/res/…`	Android resources — text (`values/`), images (`drawable/` , `mipmap/`), layouts (`layout/`).

File	Kaam
`app/src/main/assets/…`	Raw files jo app apne naam se padhta hai (yahan game ka HTML/JS).

AndroidManifest.xml mein kya-kya hota hai

- `<uses-permission>` — jaise INTERNET, CAMERA
- `<application>` — icon, label, theme
- `<activity>` — screens; `LAUNCHER` intent-filter wali activity app khulte hi chalti hai
- `android:exported="true"` — launcher activity ke liye zaroori

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4. `res/` ke andar kaunse folder hote hain

Folder	Kaam	Example
`values/`	Text, colours, styles	`strings.xml` , `colors.xml` , `themes.xml`
`drawable/`	Images / shapes	`bg_card.xml` , `logo.png`
`mipmap-*/`	App icon (har screen density ke liye alag size)	`ic_launcher.png`
`layout/`	Screen ka design (XML)	`activity_main.xml`
`xml/`	Settings files	`backup_rules.xml`
`raw/`	Media files	`beep.mp3`
`anim/`	Animations	`slide_in.xml`

Naming rule: file ka naam **chhote letters**, **underscore**, aur **number se shuru nahi** — `my_icon.png` ✔, `My-Icon.png` ✘.

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5. `assets/` — game ka ghar

File	Kaam
`assets/index.html`	App khulte hi yahi page khulta hai (abhi placeholder hai).
`assets/config/app_config.json`	Game ki settings — board ka size, sound on/off, theme. Code badalne ke bina behaviour badal sakte hain.
`assets/game/`	Aage game ka asli code (JS files, sprites, sounds) yahan aayega.

`assets` aur `res` mein farak: `res` ki files Android compile karta hai (R.java ke through), `assets` ki files jaisi hain waisi hi rehti hain aur code se path se padhi jaati hain (`file:///android_asset/index.html`).

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6. APK ke andar kya hota hai

`app-debug.apk` ek ZIP hai, uske andar:

Entry	Kya hai
`AndroidManifest.xml`	Compiled (binary) manifest
`classes.dex`	Java/Kotlin ka bytecode
`resources.arsc`	Compiled resources (strings, colours, ids)
`assets/`	Aapki raw files (game ka HTML/JS)
`res/`	Compiled resources
`lib/<abi>/*.so`	Native libraries (agar hon)
`META-INF/`	Signature / metadata

Isi wajah se APK mein aap `.java` file nahi dekhenge — woh `classes.dex` ban jaati hai.

7. File extension ka catalogue

Extension	Kya hota hai	Kahan
`.java` / `.kt`	Code (Java / Kotlin)	`app/src/main/java/`
`.xml`	Manifest, layout, values, drawable	`app/src/main/`
`.kts`	Gradle build script (Kotlin)	root, `app/`
`.properties`	Settings (key=value)	root, `gradle/wrapper/`
`.jar`	Java library / Gradle wrapper	`gradle/wrapper/`, `libs/`
`.aar`	Android library	`libs/`
`.so`	Native library	`lib/<abi>/`
`.html`, `.css`, `.js`	Web page / game	`assets/`
`.json`	Settings / data	`assets/config/`
`.png`, `.webp`, `.jpg`	Images	`res/drawable/`, `mipmap/`
`.ttf`, `.otf`	Fonts	`res/font/`, `assets/`
`.mp3`, `.ogg`, `.wav`	Sound	`res/raw/`, `assets/`
`.apk`	Installable app (build ke baad)	`app/build/outputs/apk/`
`.aab`	Play Store upload format	`app/build/outputs/bundle/`

8. Is project ki har file — ek-ek karke

#	File	Status	Kya karti hai
1	`README.md`	✔ bani hui	Project ka parichay
2	`LICENSE`	✔	MIT license

#	File	Status	Kya karti hai
3	`.gitignore`	✔	Build files ko Git se bahar rakhta hai
4	`settings.gradle.kts`	✔	Module + repositories
5	`build.gradle.kts`	✔	Android plugin version
6	`gradle.properties`	✔	Gradle settings
7	`gradlew` / `gradlew.bat`	✔	Build chalane wali script
8	`gradle/wrapper/gradle-wrapper.jar`	✔	Wrapper ka engine
9	`gradle/wrapper/gradle-wrapper.properties`	✔	Gradle version (8.11.1)
10	`app/build.gradle.kts`	✔	App ki build settings
11	`app/proguard-rules.pro`	✔	Release rules
12	`app/src/main/AndroidManifest.xml`	✔	App ki pehchaan + activity
13	`app/src/main/java/com/snakegame/app/MainActivity.java`	✔	App ka main screen (WebView)
14	`app/src/main/res/values/strings.xml`	✔	App ka naam
15	`app/src/main/res/values/colors.xml`	✔	Rang
16	`app/src/main/res/mipmap-*/ic_launcher.png`	✔	App icon (5 sizes)
17	`app/src/main/assets/index.html`	✔ (placeholder)	Abhi ka shuruaati page
18	`app/src/main/assets/config/app_config.json`	✔	Game settings
19	`app/src/main/assets/game/`	⏳ khali	Game ka code yahan aayega
20	`.github/workflows/build-apk.yml`	✔	Har push par APK build + release
21	`docs/FILES-HINDI.md`	✔	Yahi document
22	`docs/FILE-TREE.txt`	✔	Folder tree
23	`docs/ROADMAP.md`	✔	Aage ka plan

9. Naya kuch banate waqt kahan rakhein

Aap kya bana rahe hain	Kahan rakhein
Naya screen / logic	`app/src/main/java/com/snakegame/app/`
Naya text / colour	`res/values/strings.xml` , `colors.xml`
Image / icon	`res/drawable/` (icon ke liye `mipmap-*/`)
Game ka JS / HTML / sprite / sound	`assets/game/`
Settings ka data	`assets/config/app_config.json`
Naya permission	`AndroidManifest.xml`
Nayi library	`app/build.gradle.kts` ke `dependencies { }`
Naya build step	`.github/workflows/build-apk.yml`

10. Safe practices (yaad rakhein)

1. `local.properties` kabhi GitHub par na daalein — usme aapke computer ka path hota hai.
2. `build/` folder commit na karein — woh har build par dobara banta hai.
3. File ke naam chhote letters + underscore mein rakhein (`snake_food.png`), spaces aur capital letters se bachein.
4. `applicationId` ek baar tay hone ke baad na badlein — woh app ki permanent ID hoti hai.
5. APK ka size chhota rakhne ke liye images ko `.webp` mein rakhein.
6. Har badlav ke baad: `./gradlew assembleDebug` chala kar dekhein ki build pass hai.
7. Version badhate rahein: `versionCode` (number) aur `versionName` (dikhne wala) dono.

11. Build kaise chalega

```
 # local machine par
./gradlew assembleDebug
# output: app/build/outputs/apk/debug/app-debug.apk

# GitHub par
git push  →  Actions → Build APK  →  release mein APK taiyar
```

Direct download link (build ke baad):

`https://github.com/rewpal5-dotcom/Snake-Game/releases/latest/download/app-debug.apk`

Ye document project ke saath hi update hota rahega — jab bhi koi nayi file add ho, uski entry table (section 8) mein add ki jayegi.